

1.

1. overlap-save method.

i. Divided $x(n)$ into sections
 $x_1(n)$ - length N . ~~$0 \leq n \leq N-1$~~
 $x(n)$

$$X(n) = \{1, 2, 3, \dots, 10\}$$

$$h(n) = \{1, 0, -1\}$$

Let $\# \text{ lengths } \{x(n)\} = 10$, $M=3$

Let $N = 4$ (length of each section).

Overlap each input section with preceding section by $(M-1)$ samples.

⇒ Because, in N -point circular convolution, first $(M-1)$ samples obtained are incorrect, i.e. not same as linear convolution.

$\Rightarrow$ overlap $(M-1) = 3-1 = 2$ samples of each S/P section

iii

At beginning, first $(M-1)$ samples of first section are set to zero.

$$\Rightarrow x_0(n) = \{0, 0, x(0), x(1), x(2), x(3)\}$$

$$x_1(n) = \{x(2), x(3), x(4), x(5), x(6), x(7)\}$$

$$x_2(n) = \{x(6), x(7), x(8), x(9), 0, 0\}$$

$$\Rightarrow x_0(n) = \{0, 0, 1, 2, 3, 4\}$$

$$x_1(n) = \{3, 4, 5, 6, 7, 8\}$$

$$x_2(n) = \{7, 8, 9, 10, 0, 0\}$$

iv

Perform circular convolution of each section $x_r(n)$ with $h(n)$ & obtain $y_r(n)$.

$$\Rightarrow y_0(n) = x_0(n) \otimes h(n) = \{y_0(0), y_0(1), y_0(2), y_0(3), y_0(4), y_0(5)\}$$

$$\Rightarrow y_1(n) = x_1(n) \otimes h(n) = \{y_1(0), y_1(1), y_1(2), y_1(3), y_1(4), y_1(5)\}$$

$$y_2(n) = x_2(n) \otimes h(n) = \{y_2(0), y_2(1), y_2(2), y_2(3), y_2(4), y_2(5)\}$$

$$\Rightarrow y_0(n) = \{-3, -4, 1, 2, 2, 2\}$$

$$y_1(n) = \{-4, -4, 2, 2, 2, 2\}$$

$$y_2(n) = \{7, 8, 2, 2, -9, -10\}$$

v

First $(M-1)$ samples of each output section must be discarded and remaining $(N-M+1)$ samples is the desired result from linear convolution.
→ Concatenate each section to obtained desired filtered output.

$$\Rightarrow y_0(n) = \{1, 2, 2, 2\}$$

$$y_1(n) = \{2, 2, 2, 2\}$$

$$y_2(n) = \{2, 2, -9, -10\}$$

$$\Rightarrow y(n) = \{ \underset{\uparrow}{1}, 2, 2, 2, 2, 2, 2, 2, 2, 2, -9, -10 \}$$

Formula:

Padding $(M-1)$ zeros in first IP section
we obtained total IP length

$$\underline{10 + 2 = 12}$$

$$\text{Number of sections required} = \frac{[\text{total length of IP sequence}]}{[\text{no. of new samples in each section}]}$$

$$= \frac{\text{total length}}{N - M + 1} = \frac{12}{4} = \underline{\underline{3}}$$

2. Overlap-add method

ex.

$$x(n) = (n+1), \quad 0 \leq n \leq 9$$

$$h(n) = \{ \underset{\uparrow}{1}, 0, -1 \}$$

compute convolution using overlap-add method.

$$\Rightarrow \text{length } \{x(n)\} = 10.$$

$$M = 3.$$

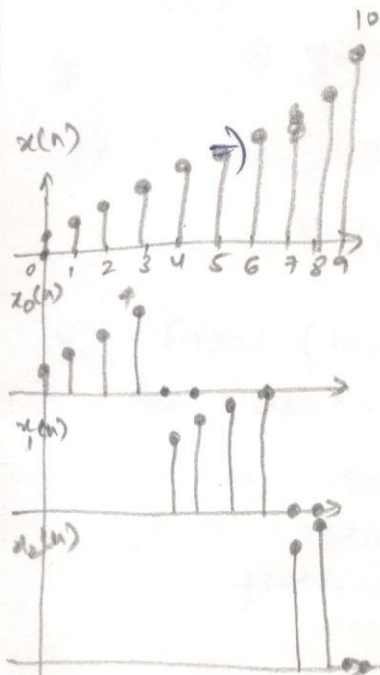
$$\text{Let } \underline{N = 4} \Rightarrow \text{length } \{x_0(n)\} = 4.$$

$\rightarrow$ Linear convolution of $x_0(n)$ & $h(n)$ will be of length $(N+M-1) = (4+3-1) = 6$.

$\rightarrow$ Thus, pad zeros in $x_0(n)$ & $h(n)$ to make them of length 6.

$$x_0(n) \Rightarrow (M-1) = 2 \text{ zeros to be padded}$$

$$h(n) \Rightarrow (N-1) = 3 \text{ zeros to be padded.}$$

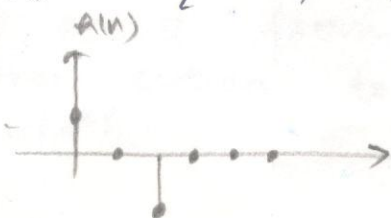


$$x_0(n) = \{ 1, 2, 3, 4, \underline{0, 0} \}$$

$$x_1(n) = \{ 5, 6, 7, 8, \underline{0, 0} \}$$

$$x_2(n) = \{ 9, 10, \underline{0, 0, 0, 0} \}$$

$$h(n) = \{ 1, 0, -1, \underline{0, 0, 0} \}$$



→ Compute linear convolution using 6-point circular convolution of each section $x(n)$ with $h(n)$.

$$y_0(n) = h(n) \otimes x_0(n)$$

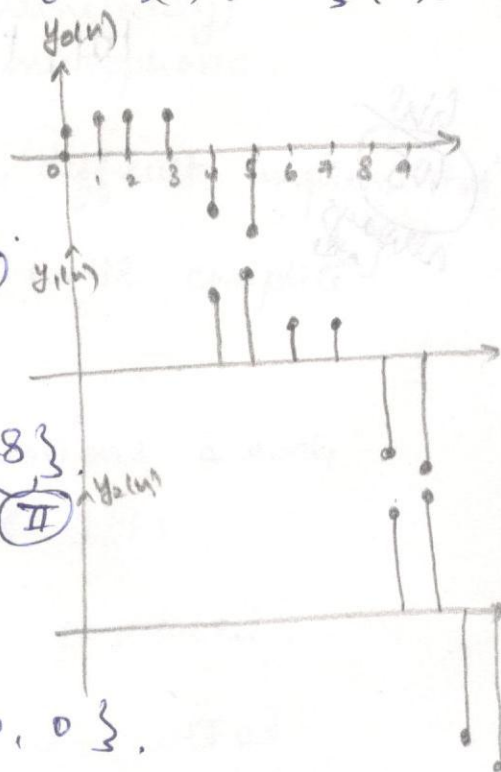
$$= \{1, 2, 2, 2, \underbrace{-3, -4}_{\text{add (I)}}\}$$

$$y_1(n) = h(n) \otimes x_1(n)$$

$$= \{ \underbrace{5, 6, 2, 2}_{\text{add (I)}}, \underbrace{-7, -8}_{\text{add (II)}} \}$$

$$y_2(n) = h(n) \otimes x_2(n)$$

$$= \{ \underbrace{9, 10, -9, -10}_{\text{add (II)}}, 0, 0 \}$$



→ Filtered o/p section will overlap by $(M-1)=2$ samples.

These overlapping samples must be added to get final result.

$$\Rightarrow y(n) = \{ \underset{\uparrow}{1}, 2, 2, 2, 2, 2, 2, 2, 2, 2, -9, -10 \}$$

Formula.

$$\text{Number of 4P sections required} = \frac{[\text{length of 4P sequence}]}{[\text{length of each section}]} = \frac{10}{4}$$

$$\approx 3.$$